### 3.3.2. Summary of glycolysis

At the end of glycolysis, one glucose molecule is converted into two pyruvate molecules. In addition we also have two ATP, two NADH molecules formed from ADP and $NAD^+$ molecules respectively. To regenerate the coenzyme $NAD^+$, NADH must be re-oxidised. If oxygen is available, the pyruvate is oxidized all the way to carbon dioxide in cellular respiration, regenerates $NAD^+$. During this process more ATP molecules are produced. Under anaerobic conditions it is converted to lactate by lactate dehydrogenase enzyme and regenerates $NAD^+$. Here no further ATP molecules are produced.

### 3.3.3. Energetics of glycolysis

During the glycolysis of one molecule of glucose, two molecules of ATP and two molecules of NADH are produced. The ATP molecules serves as energy source. The NADH molecules are oxidised in mitochondria producing more number of ATP molecules (approximately 3 ATPs per NADH molecule).

| Step No. | Description | Number of ATP/ NADH formed/ consumed |
|---|---|---|
| 6 | Formation of 1,3-bisphospho glycerate from glyceraldehyde 3-phosphate | 2 NADH |
| 7 | Formation of 3 phosphoglycerate from 1,3 bisphospho glycerate | 2 ATP |
| 10 | Formation of pyruvate from phosphoenol pyruvate | 2 ATP |
| 1,3 | Formation of Glucose-6-phosphate & Fructose-1,6-bisphosphate | 2 ATP (consumed) |
| Number of net ATP molecules generated per one molecule of glucose | | 2 ATP |
| Number of net NADH molecules generated one molecule of glucose | | 2 NADH |

### 3.3.4. Glycolysis under anaerobic condition:

Some organisms are able to continually convert energy without the presence of oxygen. They undergo glycolysis, followed by the anaerobic process to produce ATP.

As discussed earlier anaerobic glycolysis is the transformation of glucose to lactate when limited amounts of oxygen ($O_2$) exist.

Muscle cells can continue to produce ATP when oxygen runs low using lactic acid fermentation. However, this often results in muscle fatigue and pain due to accumulation of lactate. When sufficient oxygen is not present in the muscle cells, the pyruvate, the terminal electron acceptor is converted to lactate by the enzyme lactate dehydrogenase, regenerating $NAD^+$.



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